

Akash Joshi

AI/ML Engineer

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Profile

AI/ML Engineer with 4+ years of experience in designing and deploying machine learning, deep learning, and NLP solutions across diverse domains. Skilled in building scalable ML pipelines and applying advanced NLP techniques for real-world problems. Currently focused on Generative AI, developing applications with LLMs, RAG, and multi-agent systems to deliver enterprise-ready AI solutions.

Technical Skills

Language & Libraries

Python, SQL, Pandas, Numpy, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn, NLTK

Generative AI & LLMs

Hugging Face, LangChain, LangGraph, OpenAI API, Prompt Engineering, RAG, RAG Evaluation (BLEU, ROUGE), AI Agents, Large Language Models (LLMs).

Machine Learning

Regression, Classification, Clustering, EDA, Random forest, Xgboost, Hyperparameter Tuning, Cross Validation, Model Evaluation, Feature Engineering

Natural Language Processing (NLP)

Text Preprocessing, Tokenization, Stemming & Lemmatization, Embedding Techniques, Sentiment Analysis, NER, BERT.

DevOps & Tools

Docker, Git, Jupyter Notebook

Cloud Platforms

AWS (EC2, S3)

Vector Databases

Pinecone, FAISS, Chroma

Deep Learning

ANN, RNN, CNN, LSTM, GRU, Attention Mechanisms, Transformer Architectures, Transfer Learning.

Professional Experience

Asics Tech, AI/ML Engineer

07/2024 – Present

- AI/ML Engineer with experience in designing scalable machine learning pipelines and NLP-driven applications across airline, oil & gas
- Proficient in Python and SQL, with hands-on experience in AI workflow (langchain, langgraph), ML/DL frameworks such as Scikit-learn, TensorFlow and NLP libraries like spaCy and NLTK.
- Integrated Gen AI techniques, including RAG pipelines and prompt engineering with LLMs to enhance analytical outputs and automate data interpretations.
- Performed extensive data preprocessing and feature engineering, handling missing data, outliers, and feature selection to improve model accuracy and generalization.
- Conducted model evaluation and performance monitoring using metrics such as RMSE, MAE, Precision, Recall, F1-score, and implemented model retraining strategies for continuous improvement.
- Applied version control using Git to manage code changes and collaborate efficiently with team members.

eTeam Infoservices, Software Engineer

04/2021 – 06/2024

- Evaluated and monitored ML models for demand forecasting, shipment delay prediction, and ticket classification using RMSE, MAE, Precision, Recall, and F1-score, and built automated retraining pipelines to handle data drift and seasonal demand changes in e-commerce logistics
- Used Git-based branching strategies to manage model experiments, feature engineering updates, and deployment configurations across e-commerce recommendation and logistics optimization pipelines.
- Documented end-to-end workflows, including data ingestion from ERP/CRM systems, model architecture (BERT, XGBoost, LSTM)

Education

B.Tech in Computer Science & Eng, SGGS Nanded

Projects

Agent-Based RAG System for Airline Baggage Queries

- Designed and implemented a LangChain-based workflow integrating RAG with API calls to handle airline baggage queries.
- Built an LLM-powered Agent for classifying queries into knowledge, data, or hybrid types, ensuring accurate and safe routing.
- Developed a RAG pipeline over airline baggage manuals using PyMuPDFLoader, hybrid chunking, OpenAI embeddings, and Pinecone.
- Implemented metadata-aware indexing and filtering in Pinecone using attributes such as department, role, and source file, improving retrieval precision and traceability.
- Built an agent-based RAG setup to reason over retrieved baggage knowledge and generate context-aware, structured responses for scenarios such as lost, delayed, and damaged baggage.
- Applied instruction-based prompting to reduce hallucinations and improve response reliability. Chose and optimized multiple LLMs for balancing accuracy, cost, and efficiency.
- Used InMemory Saver for lightweight, session-based memory to support conversational follow-ups. Built REST APIs using FastAPI to deploy this AI Model.
- Utilized AWS services for model deployment, scalability, and monitoring.

Aware Retrieval System for Oil & Gas Safety Queries

- Designed and implemented a RAG workflow to support drilling operations and safety-related queries across drilling manuals, HSE guidelines, emergency procedures, and regulatory documents.
- Developed a RAG pipeline over oil & gas drilling and safety documentation using PyMuPDFLoader, hybrid chunking, embedding models, and FAISS.
- Implemented metadata-aware indexing and filtering in Pinecone using attributes such as operation type, rig category, safety classification, document type, section, and source file, improving retrieval relevance, traceability.
- Applied a pure RAG approach, grounding all responses strictly in retrieved drilling and safety context to ensure accuracy, explainability, and regulatory in safety-critical scenarios.
- Used instruction-based prompting to reduce hallucinations and risk-aware responses, particularly for emergency procedures and operational safety queries.
- Optimized chunk size, overlap, and semantic boundaries to handle long, complex technical documents, including drilling programs, well control manuals, and HSE standards.

Customer Retention Model for E-Commerce

- Built and interpreted model behavior using feature importance scores from XGBoost to identify key drivers influencing e-commerce customer outcomes such as purchase conversion and churn.
- Developed customer churn prediction model using XGBoost, customer demographics, browsing behavior, order history, cart activity, and transaction data.
- Performed data preprocessing and feature engineering, including handling missing values, and creating features (e.g., order/ session frequency, average order value), improving model performance.
- Conducted EDA to uncover behavioral patterns, class imbalance, and correlations between customer attributes and churn behavior.
- Applied L1 (Lasso) and L2 (Ridge) regularization within XGBoost to reduce overfitting, control model complexity, and improve generalization on unseen data.
- Evaluated model performance using multiple metrics, including accuracy, precision, recall, F1-score, and AUC-ROC.
- Reducing customer churn by an estimated 10-15% annually

Logistics Demand Forecasting & Shipment Delay Prediction System

- Designed and implemented ML models for demand forecasting and shipment delay prediction in e-commerce logistics using historical orders, delivery timelines, warehouse data, and seasonal trends.
- Built regression and classification models (XGBoost, Random Forest) to predict order demand volumes and identify high-risk delayed shipments
- Performed extensive feature engineering including lead time, route distance, carrier performance, warehouse load, holiday indicators, and seasonal patterns.
- Applied regularization and hyperparameter tuning to prevent overfitting and improve model generalization.